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ROLLNO.: 20

COURSE: AI&DS

CLASS: BE

SUB: Computer Laboratory-III

Title: Design a Distributed Application Using RPC for Remote Computation of Factorial

Code:

```
#server.py

from xmlrpc.server import SimpleXMLRPCServer
# Function to calculate factorial
def factorial(n):
    if n < 0:
        return "Factorial not defined for negative numbers"
    result = 1
    for i in range(1, n + 1):
        result *= i
    return result
# Create RPC server
server = SimpleXMLRPCServer(("localhost", 8000))
print("RPC Server is running on port 8000...")
# Register function
server.register_function(factorial, "factorial")

# Run server
server.serve_forever()

#client.py
import xmlrpc.client
# Connect to RPC server
server = xmlrpc.client.ServerProxy("http://localhost:8000/")
# Take input from user
```

```
num = int(input("Enter a number: "))  
# Call remote factorial function  
result = server.factorial(num)  
# Display result  
print("Factorial of", num, "is:", result)
```

Output:

RPC Server is running on port 8000...

Enter a number: 5

Factorial of 5 is: 120